Aristotle's Pyramid

Which face pyramids of the Platonic solids fill space: alone only the cube's, in pairs only the tetrahedron's with the octahedron's, in equal numbers; the icosahedron's, the Cleo-Icos, needs a partner that is not a face pyramid

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Version 1 · 21 October 2026 · DICTO DOI: reserved on publication · Licence CC BY-SA 4.0 · Scripts, exact checks and figures: https://github.com/DICTOR-Master/kaleidohedra/tree/0294763/studies/2026-10-21-aristotle-pyramids (research branch, commit 0294763; also attached to this record)
The question, whether Aristotle could have meant the Cleo-Icos, is DICTO's, as are the names Cleo-Icos, Antonius and Admiral Caps and the earlier findings of §5; computation, proofs and drafting with Claude (Anthropic).

Abstract. Aristotle wrote that among solids only "the pyramid and the cube" fill space (On the Heavens III.8). Read as the regular tetrahedron this is false, and the error is usually explained by near misses: five regular tetrahedra round an edge leave a gap of 7.36°, and twenty round a point leave small gaps. DICTO asked whether Aristotle could have meant another pyramid: the Cleo-Icos, the pyramid from the icosahedron's centre over one face. We take the five *face pyramids* of the Platonic solids (the cone from the centre over one face) and prove, with a Dehn-type counting map made rigorous by the theorems of Conway, Radin and Sadun on angles with rational squared cosines: in any tiling of space by copies of them (face to face or not), the dodecahedron's and the icosahedron's pyramids never occur, and the tetrahedron's occurs exactly when the octahedron's does. So alone only the cube's pyramid fills space; in pairs only the tetrahedron's and the octahedron's do, as the octet truss cut up, and in any such tiling with equal edges their numbers are asymptotically equal. The Cleo-Icos fills space only with a partner that is not a face pyramid: Antonius, the middle third of the prolate golden rhombohedron (#20). A table of DICTO's earlier exact fillings shows how each whole Platonic solid fills space with partners; for the dodecahedron and the icosahedron the partner is set by Euclid's roof or its icosahedral form. The angle identities, the counting over all 31 sets of pyramids and the two rhombohedra are checked in exact arithmetic.

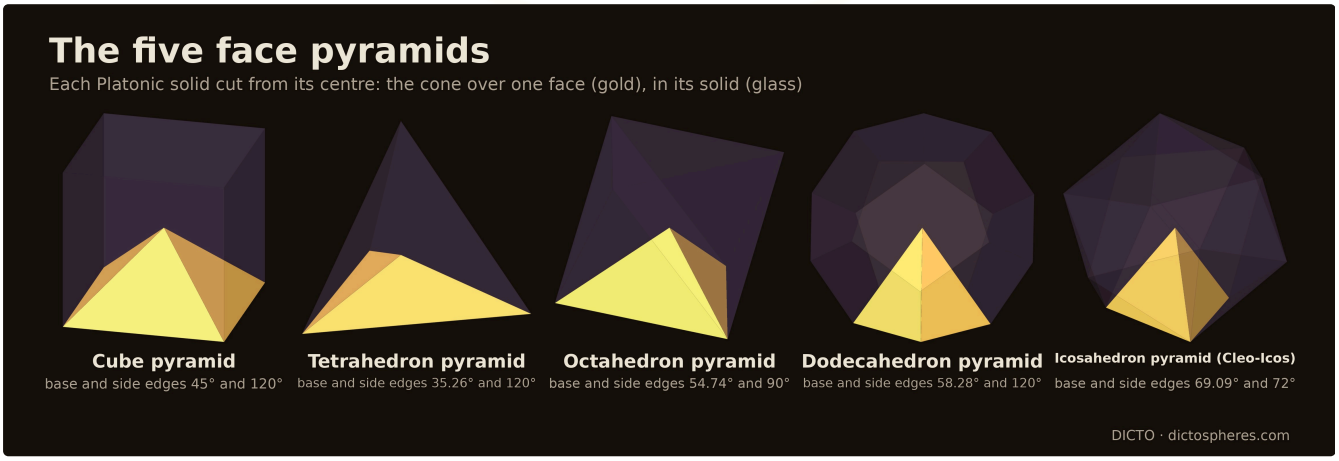


Figure 1. The five face pyramids (gold), each shown in its solid (glass) with its tip at the solid's centre and the face at the bottom: cube, tetrahedron, octahedron, dodecahedron and icosahedron (the Cleo-Icos). Under each: the dihedral angles along its base edges and its side edges (Table 1).

1. DICTO's question

"Among surfaces it is agreed that there are three figures which fill the place that contain them, the triangle, the square and the hexagon; among solids only two, the pyramid and the cube" (Aristotle, On the Heavens III.8, 306b, in Guthrie's translation as quoted by Lagarias and Zong [1]). If "the pyramid" is the regular tetrahedron, the claim is wrong: its dihedral angle $\alpha = \arccos(1/3) \approx 70.53°$ is not a whole fraction of 360°, and five tetrahedra round an edge leave 7.36° [1]. Lagarias and Zong trace the long correction [1]: Simplicius and Averroës held that twelve regular tetrahedra fill space round a point; others said twenty; Thomas Bradwardine observed that the icosahedron cuts from its centre into twenty congruent tetrahedra but could not decide whether they are regular; Paul of Middelburg pointed out that they are not; and Maurolico (1529) stated that cubes alone, and tetrahedra together with octahedra, fill space. Twenty regular tetrahedra can indeed meet at a point, with small gaps between them [1]: the near miss behind the commentators' twenty.

DICTO's question goes the other way: could Aristotle's pyramid have been Bradwardine's tetrahedron, the icosahedron's own pyramid, which DICTO calls the Cleo-Icos, reached from his CLEO piece [2, 3]? We widen it to all five **face pyramids**: for a Platonic solid with centre O, the face pyramid over a face F is the convex hull of F and O (Figure 1). Its dihedral angle along a base edge is half the solid's dihedral angle, since the plane through O and the edge bisects it; along a side edge (from O to a corner) it is 360°/q, where q faces meet at that corner. These are checked exactly in pyramid_checks.py.

Face pyramid	Base edges	Dihedral along a base edge	Side edges	Dihedral along a side edge
cube	4	45° = π/4	4	120° (3 faces at a corner)
tetrahedron	3	$\alpha/2 \approx 35.26°$	3	120°
octahedron	3	$(\pi - \alpha)/2 \approx 54.74°$	3	90° (4 faces)
dodecahedron	5	$\theta_D/2 \approx 58.28°$	5	120°
icosahedron (Cleo-Icos)	3	$\theta_I/2 \approx 69.09°$	3	72° (5 faces)

Table 1. The five face pyramids. $\alpha = \arccos(1/3)$, $\theta_D = \arccos(-1/\sqrt{5})$ and $\theta_I = \arccos(-\sqrt{5}/3)$ are the dihedral angles of the tetrahedron, dodecahedron and icosahedron; the octahedron's is $\pi - \alpha$ (its cosine is $-1/3$) and the cube's $\pi/2$.

2. The counting map

Lemma 1 (angles round an edge). Let space be tiled by convex polyhedra, finitely many meeting any bounded set, and let e be an edge of a tile P. For all but finitely many points x of e, the tiles containing x have x either in the relative interior of an edge on the line of e, or in the relative interior of a face whose plane contains that line; their dihedral angles at x add up to 2π, or to π when one tile has x inside a face (the edge lies on a face).

Proof. Only finitely many tiles meet a neighbourhood of e. Leave out of e the finitely many points that are corners of these tiles, or where an edge of one of them crosses the line of e. At any other point x of e, no tile has x in its interior (it would overlap P), and a tile with x inside an edge has that edge on the line ℓ of e. If a tile T has x inside a face, near x it is a half-space; P lies in the other closed half-space, and so does the line ℓ through x, so the plane contains ℓ. Near x every tile is therefore a wedge or a half-space bounded by planes through ℓ, and in the plane at right angles to ℓ through x they become sectors that cover a disc with disjoint

interiors: their angles add up to 2π. A half-space counts π, and there is at most one, since two would leave no room for P. ∎

No face-to-face condition is needed. Now let h be a map from the angles to the rationals that is additive with rational coefficients and has $h(\pi) = 0$. At every such point x the values of h at the dihedral angles there add up to $h(2\pi) = 0$ or $h(\pi) = 0$. Such maps exist with any values on α, θ_D and θ_I, because:

Lemma 2. π, α, θ_D and θ_I are linearly independent over the rationals.

Proof. $\cos^2 \alpha = 1/9$, $\cos^2 \theta_D = 1/5$ and $\cos^2 \theta_I = 5/9$, so $\tan \alpha = 2\sqrt{2}$, $\tan \theta_D = -2$ and $\tan \theta_I = -2/\sqrt{5}$: all three are pure geodetic angles in the sense of Conway, Radin and Sadun [4], with tangents rational multiples of √2, √1 and √5, three different square classes. None is a rational multiple of π: if $\theta = (m/n)\pi$ then $e^{i\theta}$ is a root of unity and $2 \cos \theta$ an algebraic integer, but $(2 \cos \theta)^2$ is 4/9, 4/5 and 20/9, not integers. (Theorem 2 of [4] writes each angle as a rational multiple of π plus their basis angles; this shows the basis part is not zero.) Suppose a rational combination $q_1\alpha + q_2\theta_D + q_3\theta_I$ is a rational multiple of π. By the Splitting Theorem of [4, Theorem 3], so is each part whose tangents are rational multiples of one square root: $q_1\alpha$, $q_2\theta_D$ and $q_3\theta_I$ separately; hence $q_1 = q_2 = q_3 = 0$. ∎

The angles of Table 1 in this basis: the cube pyramid's are π/4 and 2π/3, the side edges' 2π/3, π/2 and 2π/5, all rational multiples of π ($h = 0$); the base edges carry ½α (tetrahedron), ½π − ½α (octahedron), $\frac{1}{2}\theta_D$ (dodecahedron) and $\frac{1}{2}\theta_I$ (Cleo-Icos). This is the argument of #20 for the Cleo-Icos alone [3, §5.6–5.7], carried out for all five and made rigorous with [4].

3. Which face pyramids fill space

> **Theorem 1.** Take any finite set of the five face pyramids, at any fixed sizes, mirror images allowed, and any tiling of space by copies of them (not necessarily face to face). Then (a) no tile is a dodecahedron pyramid or a Cleo-Icos; (b) tetrahedron pyramids occur if and only if octahedron pyramids occur.

Proof. (a) Choose $h(\theta_D) = 1$ and $h(\pi) = h(\alpha) = h(\theta_I) = 0$. Then the base edges of a dodecahedron pyramid carry $h = ½$, and every other dihedral angle of every pyramid carries $h = 0$. At a point x of a base edge of a dodecahedron pyramid, chosen as in Lemma 1, the values of h add up to at least ½, not 0: a contradiction. The same with $h(\theta_I) = 1$ excludes the Cleo-Icos. (b) Choose $h(\alpha) = 1$, the rest 0: a tetrahedron pyramid's base edges carry +½, an octahedron pyramid's −½, all other angles 0. Without octahedron pyramids, a base edge of a tetrahedron pyramid gives a sum of at least ½; without tetrahedron pyramids, a base edge of an octahedron pyramid gives at most −½. ∎

Corollary. *Alone*, only the cube pyramid fills space (six make a cube). *In pairs* with both kinds used, only the tetrahedron and octahedron pyramids do; pairs with the cube pyramid can only tile without their second piece, and every pair with the dodecahedron pyramid or the Cleo-Icos fails. The checks run Theorem 1's test over all 31 non-empty sets of the five pyramids: only {cube}, {tetrahedron, octahedron} and {cube, tetrahedron, octahedron} pass, and with all five allowed the dodecahedron pyramid and the Cleo-Icos are still excluded.

The octet cut up. The tetrahedron and octahedron pyramids do fill space, as Maurolico's tiling [1] cut from the centres. Two regular tetrahedra on opposite faces of a regular octahedron make a rhombohedron of 60° that tiles space by the face-centred cubic lattice [1]; the checks confirm it exactly with lattice vectors (1, 1, 0), (1, 0, 1), (0, 1, 1): octahedron 4/3 + 2 tetrahedra of 1/3 = the cell, 2. Cut from their centres they give **8 octahedron pyramids and 8 tetrahedron pyramids per cell**, equal numbers (Figure 2, top), and round each edge of the octet, between 2 tetrahedra and 2 octahedra, $4 \times \alpha/2 + 4 \times (\pi - \alpha)/2 = 2\pi$.

> **Proposition 2 (equal numbers).** In any tiling of space by tetrahedron pyramids of edge a and octahedron pyramids of edge b, the numbers $N_T(R)$ and $N_O(R)$ of each inside a ball of radius R satisfy $|a\,N_T(R) - b\,N_O(R)| \le C\,R^2$ for a constant C. So their numbers are in the ratio b : a in the limit, equal when the edges are equal, as in the octet.

Proof. With $h(\alpha) = 1$, Lemma 1 gives a sum of 0 at all but finitely many points of each edge. Integrate this sum along all edges inside the ball: Σ over tiles of Σ over their edges of h(angle) × (length of the edge inside the ball) = 0. A tetrahedron pyramid wholly inside contributes $3 \times a \times \tfrac{1}{2}$, an octahedron pyramid $3 \times b \times (-\tfrac{1}{2})$; the tiles that meet the sphere are at most $C'R^2$ in number (bounded size, volume bounded below) and contribute a bounded amount each. ∎

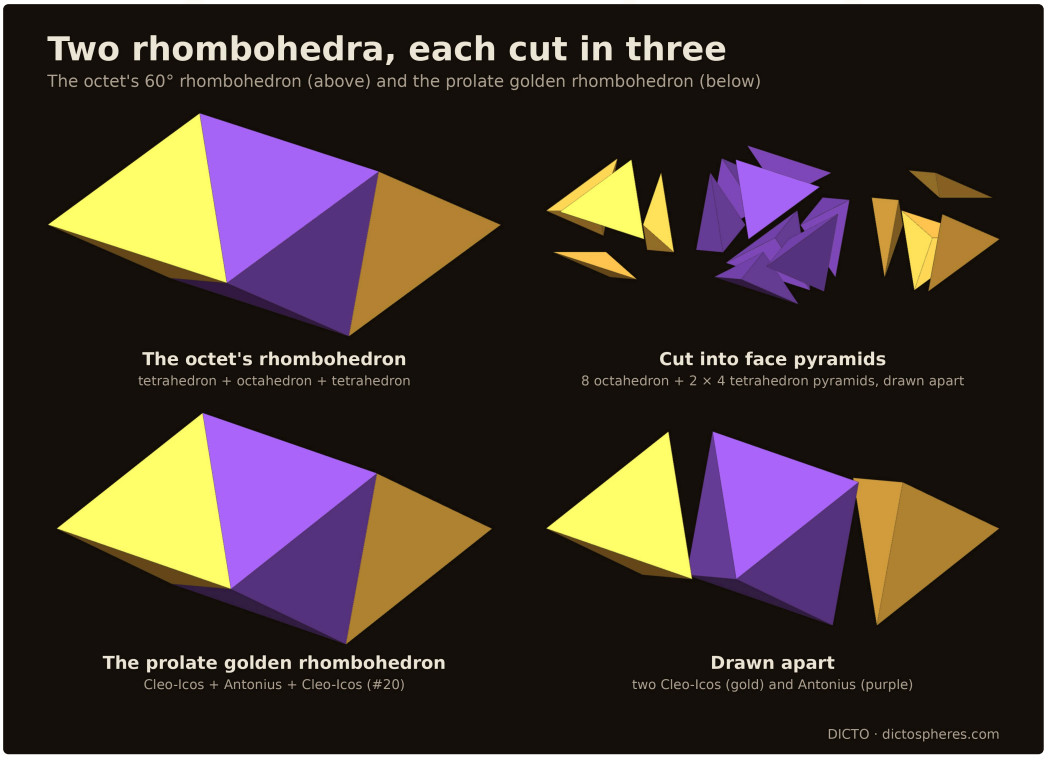


Figure 2. Above: the octet's 60° rhombohedron, tetrahedron + octahedron + tetrahedron (gold, purple, gold), whole and cut into its 8 octahedron pyramids and 2 × 4 tetrahedron pyramids, drawn apart. Below: the prolate golden rhombohedron, Cleo-Icos + Antonius + Cleo-Icos (#20), whole and drawn apart along its long axis. Each whole rhombohedron is scaled to fit its panel.

4. The Cleo-Icos needs Antonius

By Theorem 1 the Cleo-Icos fills space with none of the five face pyramids. It does fill space with one partner that is not a face pyramid [3, §5.7]: the three lines from the icosahedron's centre to the corners of a face meet at $\arccos(1/\sqrt{5}) \approx 63.43°$, the angle of the golden rhombus, so they span a prolate golden rhombohedron, and the planes $x + y + z = 1$ and 2 (in coordinates along those lines) cut it into a Cleo-Icos at each sharp corner and **Antonius** in

the middle, a trigonal antiprism of exactly 4 times the Cleo-Icos's volume (the rhombohedron is 6 times). Golden rhombohedra tile space by a lattice, so **Cleo-Icos and Antonius fill space, two to one** (Figure 2, below). The checks confirm this split exactly and give the counting map's view of it: along Antonius's 6 triangle edges the angle is $\pi - \theta_I/2 \approx 110.91°$, which with the Cleo-Icos's $\theta_I/2$ makes a flat 180°, and along its 6 side edges the rhombohedron's 72° and 108°. With $h(\theta_I) = 1$, edges weighted by length, a Cleo-Icos carries $3 \times ½$ and Antonius $6 \times (-½)$, -2 times as much, which the two-to-one ratio balances (the Dehn invariant remark of [3]). The same cut of the 60° rhombohedron is the octet's tetrahedron + octahedron + tetrahedron [3]: in both, the middle third is the partner.

5. The whole solids and their partners

DICTO asked, too, how the whole Platonic solids fill space. His earlier exact work answers it for all five:

Solid	Fills space with	Source
cube	alone	classical
tetrahedron, octahedron	each other, two tetrahedra to one octahedron (the octet truss)	classical; Maurolico (1529) [1]
dodecahedron	Dogstars: dodecahedra on alternate cubes of a cubic lattice, each a cube with six of Euclid's roofs [5]; the hole in each other cube is the Dogstar; the dodecahedra take $(5 + \sqrt{5})/8 \approx 90.45\%$ of space	tiling and Dogstar: Hart (1996) [6]; exact: The DICTO Sunstar, #42 [7]
icosahedron	DICTO's icosahedral octet: an icosahedron + 6 edge roofs (Admiral Caps) is the regular octahedron, so icosahedra (edge 1) with their roofs in the octet's octahedra, and icosahedra of edge 1/2 with corner and edge caps in its tetrahedra, fill space; the icosahedra take $(175 - 75\sqrt{5})/12 \approx 60.79\%$	exact (released finding F21; not a separate paper) [8]; the icosahedron in the octahedron at the golden section is classical (Pacioli; Fuller's Jitterbug)

Table 2. How each whole Platonic solid fills space with partners. The idea of F21 is DICTO's: "the same principle as the dodecahedron and roofs, but with six" [8].

The pattern is the same for both golden solids. Euclid builds the dodecahedron on a cube with a roof on each face [5]; the roofs are what let dodecahedra sit on a cubic lattice, and what is left is the Dogstar. DICTO moved the roofs to the icosahedron: six edge roofs make it an octahedron, which the octet already places. Theorem 1 says why such help is needed: a solid's face pyramids carry half its dihedral angle along each base edge, and only for the cube ($\pi/2$) is that angle a rational multiple of π; the tetrahedron's and the octahedron's cancel each other (α and $\pi - \alpha$), while θ_D and θ_I cancel with nothing among the five.

6. A reading of Aristotle

We do not claim to know what Aristotle meant; Lagarias and Zong take "pyramid" to be the regular tetrahedron [1], and that is the usual reading. Within the reading DICTO asked about, the pyramid as a face pyramid of a regular solid, the answer is exact: Aristotle's two figures come out right only as the cube's own pyramid (trivially, since six make the cube) and as the tetrahedron's pyramid taken together with the octahedron's, which is Maurolico's correction [1] cut from the centres. Bradwardine's twenty tetrahedra of the icosahedron, the Cleo-Icos, cannot fill space with any face pyramid; they need Antonius.

7. Prior art and what is claimed

- **Known, credited:** the history of Aristotle's claim, the near miss, Bradwardine's and Paul of Middelburg's remarks on the icosahedron's twenty tetrahedra, Maurolico's tiling and the 60° rhombohedron of two tetrahedra and an octahedron (Lagarias and Zong [1]); Dehn's angle argument and the geodetic-angle theorems (Conway, Radin and Sadun [4]); the tiling of the cube's face pyramids (six make a cube); the dodecahedron–Dogstar tiling (Hart [6]); the icosahedron in the octahedron at the golden section (Pacioli; the Jitterbug); the Cleo-Icos itself, Mackay's unit tetrahedron and a pyramid of Hypsicles (as credited in [3]).
- **DICTO's earlier work, cited:** the Cleo-Icos and Antonius filling and the counting argument for the Cleo-Icos alone (#20 [3]); the exact Sunstar Lattice (#42 [7]); the icosahedral octet (F21 [8]).
- **Claimed here:** Theorem 1, which face pyramids of the Platonic solids can occur in a tiling of space, alone, in pairs or in any set, with the proof by Lemmas 1 and 2; Proposition 2, the equal numbers; and the reading of Aristotle as a question about face pyramids, DICTO's question.

In the sources checked (the full text of the 81 library sources with a text layer listed in the appendix, searched on 21 October 2026; the library's 173 link-only entries were not searched) we did not find a classification of the tilings by the face pyramids of the Platonic solids.

Data and reproducibility

In studies/2026-10-21-aristotle-pyramids: pyramid_checks.py (exact arithmetic in $\mathbb{Q}(\sqrt{5})$ with Python fractions; angles handled through $\cos^2$ and the sign of cos; log pyramid-checks.log, 45 PASS, no FAIL): the solids' dihedral angles; every base and side angle of the five pyramids; the facts used in Lemma 2; the counting test over all 31 sets; the octet cell and its 8 + 8 pyramids; the golden rhombohedron, Cleo-Icos, Antonius and their angles. angle_relations.py (angle-relations.log): the angles to 80 digits and a PSLQ search finding no integer relation among π, α, θ_D, θ_I, evidence only, the proof being Lemma 2. Figures: aristotle_figs.py (float copies of the exact solids into the viewer of studies/2026-10-19-five-fold-octet), jewel-shots.mjs (Playwright, 2× pixel ratio) and aristotle_compose.py. The 2,187-piece patch of Cleo-Icos and Antonius is that of [3].

References

1. J. C. Lagarias, C. Zong, Mysteries in packing regular tetrahedra, Notices Amer. Math. Soc. 59 (2012) 1540–1549, doi:10.1090/noti918 (quoting Aristotle, On the Heavens, tr. W. K. C. Guthrie, Loeb Classical Library 338, 1939, 306b).
2. DICTO, AXE, CLEO and FUJI, #19, DICTO DOI: 10.5281/zenodo.23351291.
3. DICTO, The FUJI Family, #20, DICTO DOI: 10.5281/zenodo.23357896 (§5.6 the Cleo-Icos, §5.7 Cleo-Icos and Antonius fill space).
4. J. H. Conway, C. Radin, L. Sadun, On angles whose squared trigonometric functions are rational, Discrete Comput. Geom. 22 (1999) 321–332; arXiv:math-ph/9812019 (Theorems 1–3).
5. Euclid, The Thirteen Books of the Elements, tr. T. L. Heath, vol. III (Books X–XIII), Cambridge (1908), Book XIII, Proposition 17.
6. G. W. Hart, Tetrahedral stellations of the dodecahedron (1996), https://www.georgehart.com/virtual-polyhedra/stellations-dodecahedron-tetrahedral.html (stellation 8).
7. DICTO, The DICTO Sunstar, #42, DICTO DOI: 10.5281/zenodo.23497702.
8. DICTO, Kaleidohedra by DICTO, release v2026.10.11-octet, finding F21 "The icosahedral octet", DICTO DOI: 10.5281/zenodo.23284390.

Appendix. The reference library searched

The library's manifests list 267 entries, of which 94 have a local file; these give the 81 distinct sources with a text layer listed below and 13 page images. The full text of the 81 sources was searched on 21 October 2026 for: Aristotle, De Caelo, On the Heavens, pyramid, fill space, space-filling, tile space, Dehn, dihedral angle, central pyramid, face pyramid, Maurolico, octet, tetrahedral-octahedral, golden rhombohedron, Cleo and Antonius. The relevant hits are Lagarias and Zong [1] (the history above), Conway, Radin and Sadun [4], the Dehn invariants of icosahedral tiles in Ogievetsky and Papadopolos, the tetrahedral-octahedral honeycomb (Wikipedia), and golden rhombohedra in Senechal, Dietl and Eschenburg and Kabai's demonstrations (all in the list below); none treats the face pyramids of the Platonic solids as tiles. The 13 page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, and Fedorov 1885) and the 173 link-only entries were not searched.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
E. S. Fedorov (chapter 13, English translation)	1885	Elements of the Theory of Figures, chapter 13
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
J. H. Conway, C. Radin, L. Sadun	1999	On angles whose squared trigonometric functions are rational
R. V. Moody	2000	Model sets: a survey
I. Hafner, T. Zitko	2002	Rhombic polyhedra: from dissection of the cube to space filling with prolate rhombohedra and rhombic dodecahedra of the second kind
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2007-2015	Wolfram Demonstrations: 54-Faced Space-Filling Polyhedron (2007), Golden Octet Truss (2007), A Rhombic Dodecahedron Made of Rhombic Triacontahedra (2008), Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (2008), Cluster of 14 Rhombic Triacontahedra (2015)
K. Bezdek, Z. Lángi, M. Naszódi, P. Papez	2007	Ball-polyhedra
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations

J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
L. Montejano, E. Roldán-Pensado	2017	Meissner polyhedra
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
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